ALFIDO TECH – DATA ANALYTICS INTERNSHIP

TASK 2: SALES PERFORMANCE ANALYSIS

Submitted by: Mirsaji S | **Tools Used:** Python, Pandas, NumPy, Matplotlib, Seaborn | **Environment:** Google Colab

PAGE 1: PROJECT OVERVIEW

1. Objective

The objective of this project is to analyze Superstore sales data to understand sales performance, profitability, customer segments, and monthly sales trends. The analysis helps identify business insights and support data-driven decision-making.

2. Dataset Overview

The dataset contains sales transactions, product categories, customer details, sales amounts, and profit information. Python was used to analyze the data and create visualizations.

3. Executive Summary

This project analyzes Superstore sales data using Python and data visualization techniques. The analysis focuses on sales performance, profit contribution, customer segments, monthly trends, and the relationship between discounts and profit.

The reported results show total sales of approximately $2,297,201, total profit of $286,397, and 5,009 orders. The analysis helps identify opportunities to improve sales and profitability.

PAGE 2: ANALYSIS AND VISUALIZATIONS

4. Total Sales by Product Category

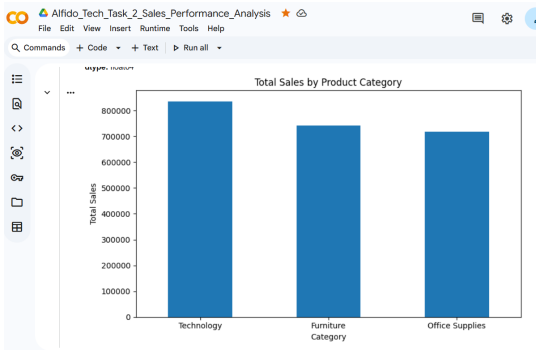


Figure 1: Total Sales by Product Category

This chart compares sales across product categories and helps identify the categories contributing most to total revenue.

5. Total Profit by Product Category

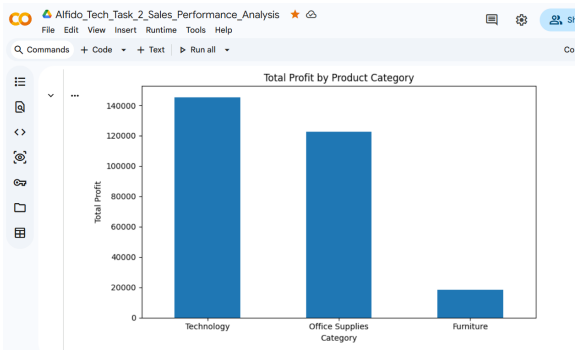


Figure 2: Total Profit by Product Category

This chart compares profit across product categories and helps identify which categories contribute most to profitability.

6. Monthly Sales Trend

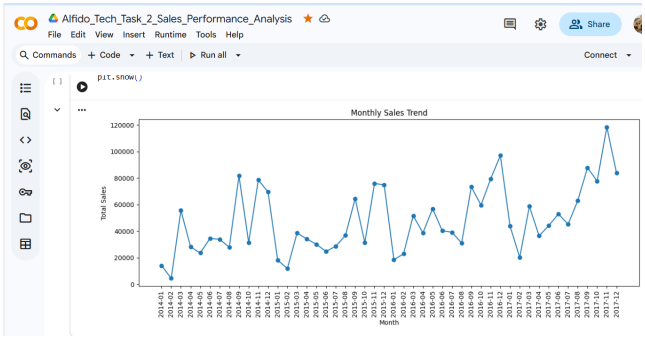


Figure 3: Monthly Sales Trend

This chart shows changes in sales over time and helps identify high-sales and low-sales periods.

PAGE 3: CUSTOMER ANALYSIS AND DISCOUNT RELATIONSHIP

7. Total Sales by Customer Segment



Figure 4: Total Sales by Customer Segment

This chart compares sales across customer segments, such as Consumer, Corporate, and Home Office. It helps businesses understand customer contributions to total sales.

8. Relationship Between Discount and Profit



Figure 5: Relationship Between Discount and Profit

This chart examines the relationship between discount levels and profit. It helps investigate whether higher discounts are associated with lower profits.

PAGE 4: KEY FINDINGS AND CONCLUSION

9. Key Findings

1. Total sales were approximately $2,297,201.
2. Total profit was approximately $286,397.
3. The dataset contained 5,009 orders.
4. Technology was identified as the highest-performing category in the reported results.
5. The charts help evaluate product category performance, customer segments, monthly sales trends, and discount-related profitability.

10. Actionable Insights and Recommendations

1. **Promote High-Performing Categories:** Focus marketing efforts on categories with strong sales and profitability.
2. **Improve Customer Marketing:** Use customer-segment analysis to develop targeted promotions and improve customer engagement.
3. **Review Discount Strategies:** Monitor the relationship between discounts and profit to support better pricing decisions.

11. Conclusion

This project uses Python and data visualization to analyze Superstore sales performance. The findings help businesses understand sales trends, evaluate product categories, analyze customer segments, and investigate the relationship between discounts and profit. These insights support better marketing and business decisions.

12. Notebook Link

Task 2 – Sales Performance Analysis Notebook:
https://github.com/mirsaji01-star/Alfido_Tech_Data_Analytics/blob/main/Task_2_Sales_Performance_Analysis/Task_2_Notebook.ipynb

Internship Organization: Alfido Tech | Domain: Data Analytics | End of Report